# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** the sunlight leaf | Starch made. |
| 2 | **(b)** 80° | i + r = 40 + 40. |
| 3 | **(b)** have continuous daylight | No change in facing. |
| 4 | **(b)** transpiration | Water vapour condenses. |
| 5 | **(d)** 20 m | Ratio 2:1. |
| 6 | **(b)** evening sky (west) | Soon after sunset. |
| 7 | **(a)** taking food from its host | Parasite. |
| 8 | **(c)** half | Inversely. |
| 9 | **(a)** umbra | Total. |
| 10 | **(a)** remain brown | No chlorophyll. |
| 11 | **(c)** 65° | 90 - 25. |
| 12 | **(a)** no seasons | Same sunlight all year. |
| 13 | **(d)** guard cells | Guard cells swell. |
| 14 | **(c)** noon | Sun overhead. |
| 15 | **(b)** sunset | Opposite the Sun. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | More light for photosynthesis (1); Greater food production (1). | 1+1 |
| 17 | Reflection 35° (1); Between rays 70° (1). | 1+1 |
| 18 | Axis is tilted (1); Sun above horizon for different durations (1). | 1+1 |
| 19 | Less water is lost in sunlight (1); Gas exchange continues (1). | 1+1 |
| 20 | It reflects more light (1); Image is bright (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Plant in a closed jar with KOH (1); Control with water (1); Test leaves for starch: only control positive (1). | 1+1+1 |
| 22 | Incidence = 90 - 70 = 20° (1); Reflection 20° (1); Between rays 40° (1). | 1+1+1 |
| 23 | New moon aligned with Sun and Earth (1); Moon's umbra falls on Earth (1); Region in umbra sees total eclipse (1). | 1+1+1 |
| 24 | Guard cells swell and the pore opens (1); They shrink and it closes (1); Light and water matter (1). | 1+1+1 |
| | **OR:** Shows the road ahead (1); Helps avoid accidents (1); Wide view (1). | 1+1+1 |
| 25 | Near the poles (1); Sun above horizon at midnight (1); During summer (1). | 1+1+1 |
| | **OR:** Hydrilla under a funnel in water (1); Test tube over stem (1); Gas collects; glowing splinter relights (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Broad flat surface (1); Thin (1); Stomata for gases (1); Veins for transport (1); Chloroplasts with chlorophyll (1). | 1+1+1+1+1 |
| | **OR:** Normal drawn (1); Angle of incidence 30° (1); Angle of reflection 30° (1); Angle between incident and reflected 60° (1); Reflected ray on opposite side of the normal (1). | 1+1+1+1+1 |
| 27 | 21 June: north summer (1); 22 Dec: north winter (1); Equinoxes: equal day and night (1); South opposite (1); Tilt same direction (1). | 1+1+1+1+1 |
| | **OR:** Water the plant (1); Cover branch with a clear bag (1); Seal the stem (1); Place in sunlight (1); Droplets inside show transpiration (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** dead wood | Saprotroph. |
| 2 | **(d)** away from the hole | Longer distance. |
| 3 | **(d)** penumbra | Partial shadow. |
| 4 | **(b)** remains brown | No light. |
| 5 | **(d)** 0° | Retraces its path. |
| 6 | **(a)** northern | North tilted to Sun. |
| 7 | **(c)** water and minerals | Large area. |
| 8 | **(c)** sunrise or sunset | Sun low. |
| 9 | **(c)** sunrise | Along with the Sun. |
| 10 | **(d)** trapping insects | Gets nitrogen. |
| 11 | **(d)** blurred and brighter | Wide beam. |
| 12 | **(c)** the Earth's shadow is large | Larger shadow. |
| 13 | **(d)** no starch is formed | No CO2. |
| 14 | **(a)** 2 m | 1 m from the mirror, so 2 m. |
| 15 | **(a)** summer | Opposite. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | No air or water (1); No weathering (1). | 1+1 |
| 17 | Rhizobium fixes nitrogen (1); Soil fertility improves (1). | 1+1 |
| 18 | Two reflections (1); Image remains upright (1). | 1+1 |
| 19 | Rays slant and spread (1); Shorter days (1). | 1+1 |
| 20 | No photosynthesis (1); Stored starch is used up (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Regular reflection in mirror (1); Diffused in wall (1); Rough surface scatters (1). | 1+1+1 |
| 22 | Lower gravity (1); One-sixth (1); Easier (1). | 1+1+1 |
| 23 | Water goes up via xylem (1); Food moves via phloem (1); Direction differs (1). | 1+1+1 |
| 24 | Height 1.4 m (1); Image 3 m behind the mirror (1); Distance from her 6 m (1). | 1+1+1 |
| | **OR:** Lamp as Sun (1); Ball moved around (1); Lit part visible changes (1). | 1+1+1 |
| 25 | More light at noon (1); More photosynthesis (1); Less light at dawn (1). | 1+1+1 |
| | **OR:** Smooth surface gives regular reflection (1); Ripples scatter light (1); Image is distorted (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Solar: Moon between Sun and Earth (1); Lunar: Earth between (1); Umbra and penumbra (1); Straight travel of light (1); Occurs rarely (1). | 1+1+1+1+1 |
| | **OR:** Rotation causes bulge (1); Equator bulges (1); Poles flatten (1); Effect small (1); Rotation important (1). | 1+1+1+1+1 |
| 27 | Potted plant kept in dark 48 h (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Test with iodine (1); Only uncovered part blue-black (1). | 1+1+1+1+1 |
| | **OR:** Closer object gives larger image (1); Farther object gives smaller (1); Larger screen distance gives larger image (1); Brightness decreases (1); Inverted always (1). | 1+1+1+1+1 |

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** cells lose turgidity | Water loss. |
| 2 | **(a)** east | Sun in the west. |
| 3 | **(d)** half disc, right half lit (northern hemisphere) | Quarter of the way. |
| 4 | **(b)** makes food by photosynthesis | Provides food. |
| 5 | **(d)** the hole is covered | No light enters. |
| 6 | **(a)** dark reddish | Sunlight bends through atmosphere. |
| 7 | **(d)** more bubbles in sunlight | Photosynthesis. |
| 8 | **(a)** 1.2 m tall | Same size. |
| 9 | **(d)** 12 hours | Equal. |
| 10 | **(b)** reduce water loss | Conserve water. |
| 11 | **(d)** decreases | Fewer rays blocked. |
| 12 | **(b)** the Sun always shines on half of it | One side faces Sun. |
| 13 | **(d)** gives shelter and absorbs water | Protection. |
| 14 | **(b)** real image | Light actually reaches. |
| 15 | **(a)** lunar | Only lunar. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | New distance 1 m from mirror (1); Image 1 m behind: 2 m apart (1). | 1+1 |
| 17 | Sun's rays are always nearly direct (1); Day length is nearly constant (1). | 1+1 |
| 18 | It takes food from the host (1); It does not make food (1). | 1+1 |
| 19 | Compare shadows of tree and stick (1); Ratio gives the height (1). | 1+1 |
| 20 | Time between two full moons (1); About 29.5 days (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Kharif in rains (1); Rabi in winter (1); Temperature and water matter (1). | 1+1+1 |
| 22 | Rhizobium fixes nitrogen (1); Soil enriched (1); Saves cost (1). | 1+1+1 |
| 23 | Look above water (1); While submerged (1); Two mirrors (1). | 1+1+1 |
| 24 | The Moon blocks the Sun's light (1); Umbra falls on the place (1); Day looks like dusk (1). | 1+1+1 |
| | **OR:** Leaf forms pitcher (1); Slippery rim and lid (1); Digestive juices (1). | 1+1+1 |
| 25 | Box with a hole and screen (1); Candle in front (1); Inverted image on the screen (1). | 1+1+1 |
| | **OR:** June longer (1); December shorter (1); Axis tilt (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | New moon (1); Waxing crescent, first quarter (1); Waxing gibbous, full moon (1); Waning gibbous, last quarter (1); Waning crescent (1). | 1+1+1+1+1 |
| | **OR:** Lichen: alga and fungus (1); Alga food, fungus shelter (1); Rhizobium and legumes (1); Plant gives sugar (1); Bacteria fix nitrogen (1). | 1+1+1+1+1 |
| 27 | Image moves towards the mirror at 1 m/s (1); Relative speed 2 m/s (1); Distances equal (1); Both distances shrink (1); So 2 m/s (1). | 1+1+1+1+1 |
| | **OR:** Night light (1); Lunar calendar (1); Festivals (1); Moon missions (1); Phases teach geometry of light (1). | 1+1+1+1+1 |

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